

Yasmine Mustafa, PhD

Yasmine Mustafa Hussain Ahmed Attia

Richland, Wa | ✉ yam64@mst.edu | ☎ (573) 537-0679 | [in](#) | [G](#)

Summary

Computer scientist and AI researcher with expertise in machine learning, multimodal learning, computer vision, and interpretable artificial intelligence. Ph.D. in Computer Science from Missouri University of Science and Technology with research focused on multimodal healthcare AI using neuroimaging, clinical, and longitudinal data. Experienced in teaching computer science courses, mentoring students, and conducting interdisciplinary research in machine learning and data-driven modeling. Interested in developing student-centered learning environments while advancing research in trustworthy and applied artificial intelligence.

Education

Missouri University of Science and Technology, PhD in Computer Science — Jan 2023 – May 2026
Rolla, MO

Dissertation focus: Anatomically-Informed Interpretable Multimodal Learning for Alzheimer's Disease Diagnosis and Progression

- Designed transformer-based architectures for multimodal healthcare data, integrating MRI, PET, and clinical narratives generated from structured patient metadata.
- Built multimodal models under missing-data conditions to reflect real-world clinical settings with incomplete or asynchronous modalities.
- Modeled disease progression using diffusion-based generative models conditioned on genetic risk factors (e.g., APOE), capturing longitudinal patterns of neurodegeneration.
- Designed attention mechanisms to align model focus with clinically relevant regions and biomarkers, improving interpretability.
- Validated models against known clinical patterns and biomarkers, and conducted ablation and attention analyses for robust evaluation.

German University in Cairo, M.Sc. in Digital Media Engineering and Technology — Mar 2021 – Jan 2022
Cairo, Egypt

Thesis: Brain-Computer Interface for Augmented Reality | Grade: A+

- Collected and processed EEG data for real-time brain-computer interface applications.
- Developed ensemble models for decoding SSVEP signals and predicting user intent from neural data.
- Analyzed neural responses to visual stimuli for attention modeling and signal characterization.

German University in Cairo, B.Sc. in Digital Media Engineering and Technology — Sept 2015 – Jul 2020
Cairo, Egypt

GPA: A+ | Semester Abroad: GUC Berlin, Germany (Sept 2019 – Jan 2020), GPA: A+

Bachelor Thesis Abroad: Fraunhofer IGD, Rostock, Germany (Mar 2019 – Aug 2019)

Thesis: Design and Development of a Telemetric Monitoring System for Joint Mobility of Knee or Hip using Machine Learning | Grade: A+

Research Experience

Multimodal AI for Behavioral Health — Missouri S&T & Saint Louis University Jan 2026 – Present

- Developed multimodal AI models integrating medical imaging and polygenic risk scores for suicide risk prediction.
- Built predictive models combining imaging-derived features and genetic risk factors under real-world, incomplete data conditions.
- Designed interpretable models in collaboration with domain experts to ensure clinical relevance and responsible use.

Research Collaborator — Tellegacy

Sep 2025 – Present

- Evaluated digital health platforms for aging-in-place care to analyze the impact of behavioral patterns on cognitive decline.
- Modeled relationships between user behavior and cognitive trajectories using longitudinal data.
- Collaborated with multidisciplinary teams to translate behavioral insights into data-driven strategies for cognitive health support.

Experience

Teaching Assistant, Washington State University — Washington, USA

Feb 2026 – May 2026

- Assisted in teaching cybersecurity and web development courses.
- Supported instruction in cybersecurity and web development courses through student mentoring, technical guidance, grading, and classroom support.
- Proctored exams and supported course administration.
- Assisted students in programming, debugging, and problem-solving during office hours and course activities.

Teaching Assistant, Missouri University of Science and Technology — Missouri, USA

Feb 2026 – May 2026

- Graded research reports and technical assignments in computer science courses.
- Provided detailed feedback on experimental design, analysis, and technical writing.
- Assisted students in understanding research methodologies and evaluation practices.
- Supported course instruction and academic activities as needed.

Teaching Assistant, German University in Cairo — Cairo, Egypt

Mar 2021 – Jan 2022

- Assisted in teaching Data Structures and Algorithms and Introduction to Computer Programming (Java).
- Guided undergraduate students through programming assignments, debugging exercises, and algorithmic problem solving.
- Supported laboratory instruction and evaluated coursework and programming projects.
- Mentored students in foundational programming concepts and software development practices.

Full-Time Software Engineer, Everteam Intalio — Cairo, Egypt

Feb 2021 – Mar 2021

- Developed and maintained enterprise software components using .NET.
- Improved system performance, debugging efficiency, and reliability.

Part-Time Full Stack Developer, Lirten Solutions — Cairo, Egypt

Aug 2020 – Jan 2021

- Built web applications using React, Node.js, MongoDB, and PostgreSQL.
- Delivered features under tight timelines while adapting quickly to new tools.

Internships

Deep Learning Intern, ARATronics Lab (Robotics) — Cairo, Egypt

Jul 2018

- Developed a deep learning model to control a prosthetic arm using Myo armband signals (MATLAB, Neural Networks Toolbox).

Web Developer Intern, German University in Cairo — Cairo, Egypt

Jun 2018

- Developed a website for the Human++ research cluster using WordPress, PHP, and JavaScript frameworks.

Speech Recognition Intern, German University in Cairo — Cairo, Egypt

Jul 2017

- Collected and processed audio data for speech recognition experiments.

Teaching Interests

Introduction to Programming, Data Structures and Algorithms, Machine Learning, Artificial Intelligence, Deep Learning, Data Science, Computer Vision, Software Engineering, Database Systems, Human-Computer Interaction, and Healthcare AI.

Research Interests

Machine learning, multimodal artificial intelligence, interpretable and trustworthy AI, computer vision, representation learning, large language models, healthcare AI, human-centered AI systems, longitudinal data modeling, and data-driven approaches for real-world decision support systems.

Professional Service

- Invited Reviewer, International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI) — reviewed 10+ submissions across multiple years; recognized on official reviewer list.
- Invited Reviewer, Association for the Advancement of Artificial Intelligence (AAAI) — reviewed 5+ submissions.
- Contributed detailed technical reviews focusing on experimental design, model validity, reproducibility, and clarity of presentation.

Technologies

Machine Learning & AI: Transformers, Vision Transformers (ViTs), foundation models, large language models (LLMs), multimodal learning, PyTorch, MONAI, HuggingFace Transformers

Healthcare Data & Modeling: Multimodal data integration (imaging, clinical, genomic), longitudinal data analysis, signal processing, feature extraction, data preprocessing, statistical evaluation

Programming: Python, C++, Java, SQL, MATLAB

Systems & Computing: Linux, Docker, Git, GPU computing (CUDA), distributed training (DDP), GPU computing, distributed training, scalable deep learning systems

Clinical AI Applications: Neuroimaging (MRI, PET), EEG signal analysis, brain-computer interfaces (BCI), clinical data modeling

Awards

Best Paper Award — PAKDD 2025 Workshop on Pattern Mining and Machine Learning for Bioinformatics 2025

- Awarded for "*Unlocking Neural Transparency: Jacobian Maps for Explainable AI in Alzheimer's Detection*".

John W. Claypool Fund for Medical Research Fellowship — Missouri University of Science and Technology 2024

- Competitive fellowship supporting research in applied artificial intelligence.

First Place — 3 Minute Thesis (3MT) — Missouri University of Science and Technology 2024

- First place in campus-wide competition; advanced to regional MAGS 3MT.

Publications

1. **Yasmine Mustafa**, Haojie Zhu, Xiaowei Yu, and Seung-Jong Park. *Anatomical Attention Transformer for Region-Wise Fusion of 3D Neuroimaging and Clinical Text in Alzheimer's Disease*. Under review, International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI), 2026.
2. **Yasmine Mustafa**, Xiaowei Yu, and Seung-Jong Park. *Anatomically Informed and Compressed 3D Vision Transformer for Dementia Detection*. Under review, BMC Bioinformatics, 2026.
3. **Yasmine Mustafa**, Haojie Zhu, Xiaowei Yu, and Seung-Jong Park. *Forecast-Guided Multimodal Fusion via Latent Transition Modeling for Alzheimer's Disease Progression*. Under review, ACM International Conference on Multimedia (ACM Multimedia), 2026.
4. **Yasmine Mustafa**, M. Elmahallawy, and T. Luo. *Unlocking Neural Transparency: Jacobian Maps for Explainable AI in Alzheimer's Detection*. Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD), 2025.
5. **Yasmine Mustafa**, M. Elmahallawy, and T. Luo. *Efficient Brain Imaging Analysis for Alzheimer's and Dementia Detection Using Convolution-Derivative Operations*. IEEE International Conference on Big Data (BigData), 2024, pp. 6420–6429.
6. **Yasmine Mustafa**, and T. Luo. *Unmasking Dementia Detection by Masking Input Gradients: A JSM Approach to*

Model Interpretability and Precision. Pacific-Asia Conference on Knowledge Discovery and Data Mining (PAKDD), 2024, pp. 75–90.

7. **Yasmine Mustafa**, and T. Luo. *Diagnosing Alzheimer’s Disease using Early-Late Multimodal Data Fusion with Jacobian Maps*. IEEE International Conference on E-health Networking, Applications and Services (Healthcom), 2023.
8. **Yasmine Mustafa**, M. Elmahallawy, T. Luo, and S. Eldawlatly. *A Brain-Computer Interface Augmented Reality Framework with Auto-Adaptive SSVEP Recognition*. Metrology for eXtended Reality, Artificial Intelligence and Neural Engineering (MetroXRINE), 2023.
9. **Yasmine Mustafa**, and S. Eldawlatly. *An Ensemble Classification Approach for Recognizing Steady-State Visually Evoked Potentials Frequencies*. IEEE International Conference on Computer Engineering and Systems (ICCES), 2021, pp. 1–6.